



Norm.

Prop. Let $f \in C([0,1])$. Then,

$$\lim_{n \rightarrow \infty} \left[\int_0^1 |f(x)|^n dx \right]^{\frac{1}{n}} = \max\{|f(x)| \mid x \in [0,1]\}.$$

Proof. Since $[0,1]$ is compact, the image of $|f|$ is also compact, so for some $a \in [0,1]$, $|f(a)| = \max\{|f(x)| \mid x \in [0,1]\}$. Observe that

$$|f(a)| = (|f(a)|^n)^{\frac{1}{n}} = \left[\int_0^1 |f(a)|^n dx \right]^{\frac{1}{n}}$$

Meanwhile, since $[0,1]$ is compact, so f is continuous uniformly on $[0,1]$.

Given $\epsilon > 0$, choose $\delta > 0$ such that $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$.

By the Minkowski inequality, for each $n \in \mathbb{N}$,

$$\left| \left[\int_0^1 |f(a)|^n dx \right]^{\frac{1}{n}} - \left[\int_0^1 |f(x)|^n dx \right]^{\frac{1}{n}} \right| \leq \left[\int_0^1 |f(x) - f(a)|^n dx \right]^{\frac{1}{n}}.$$

$$\|x+y\| \leq \|x\| + \|y\|$$

$$\|x+y\| - \|x\| \leq \|y\|$$

$$\lim_{n \rightarrow \infty} \left(\underbrace{\|x_1 - x_2 + x_2\|}_x - \underbrace{\|x_2\|}_y \right) \leq \underbrace{\|x_1 - x_2\|}_y$$

Therefore, for each $n \in \mathbb{N}$,

Prop. Let $(x_1, \dots, x_n) \in \mathbb{R}^n$ be given. Then

$$\lim_{p \rightarrow \infty} \left[\sum_{i=1}^n |x_i|^p \right]^{\frac{1}{p}} = \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

Proof. Observe that let $1 \leq j \leq n$ be the index such that

$$|x_j| = \max\{|x_1|, \dots, |x_n|\}. \quad \text{Put } (y_i)_{i=1}^n = (x_j, x_j, \dots, x_j).$$

Then $(y_i)_{i=1}^n$ satisfies that

$$\|(y_i)\|_p = \left[\sum_{i=1}^n |x_j|^p \right]^{\frac{1}{p}} = (n \cdot |x_j|^p)^{\frac{1}{p}} = \sqrt[p]{n} \cdot |x_j| \rightarrow |x_j| \text{ as } p \rightarrow \infty.$$

Now, by the Minkowski Inequality,

$$\|(y_i)\|_p - \|(x_i)\|_p \leq \|(y_i) - (x_i)\|_p = \|(x_j - x_i)_{i=1}^n\|_p = \left[\sum_{i=1}^n |x_j - x_i|^p \right]^{\frac{1}{p}}$$

